

RESEARCH ARTICLE

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Effect of pyrolysis operating conditions on the catalytic co-pyrolysis of low-density polyethylene and polyethylene terephthalate with zeolite catalysts

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Abstract

In this study, the catalytic (co-)pyrolysis of low-density polyethylene (LDPE) and polyethylene terephthalate (PET) with HZSM-5 and HY zeolite catalysts was conducted in a micro-pyrolysis reactor coupled to a two-dimensional gas chromatography system. Pyrolysis operating conditions, such as the pyrolysis temperature, the catalyst to feedstock (CF) ratio, and the LDPE:PET ratio, were varied. It was found that for the co-pyrolysis of LDPE and PET, HZSM-5 led to higher yields of C2-C4 olefins and monoaromatic products. Lower CF ratios increased the yield of C2-C4 olefins for LDPE pyrolysis, but decreased benzene yield for PET pyrolysis, concomitant with an increased yield in benzoic acid. A lower temperature of 400°C which was sufficient for the pyrolysis of LDPE, led to incomplete conversion of PET. Surface response diagrams were used to visualize the impact of the various pyrolysis operating conditions on the yield of C2-C4 olefins and BTEX, which serve as target products for the circular economy.

KEYWORDS

catalytic pyrolysis, micro-pyrolysis, mixed plastic waste, synergistic effects, zeolites

1 | INTRODUCTION

Plastic waste is a global issue facing our planet today. While plastics are extremely useful in packaging, construction materials, electronics, and many other applications, their disposable nature and widespread use has brought environmental concerns such as the great pacific garbage patch, and the leaching of microplastics into our soil and

waterways. In 2018, a total of 35 million metric tons of plastic waste was produced in the United States, with only 9% being recycled.¹ One reason for low recycling rates is due to the challenges associated with the sorting of plastic waste types, and the inability for conventional recycling methods such as mechanical recycling to process mixed plastic waste.² Multilayer packaging is one example of an inherently mixed plastic waste which often contains layers of both polyolefin polymers such as low-density polyethylene (LDPE), and polyesters such as polyethylene terephthalate (PET). One hundred million tons of multilayer thermoplastics are produced globally each year and unless their layers are deconstructed by use of solvents, they are not able to be mechanically recycled.³ The techniques available to recycle multilayer packing are limited and include using solvent targeted recovery and

Abbreviations: BTEX, benzene, toluene, ethylbenzene, and xylene; C2-C4 O, C2 to C4 Olefins; C2-C4 P, C2 to C4 Paraffins; CF ratio, catalyst to feedstock ratio; FID, flame ionization detector; GC × GC, two-dimensional gas chromatography; LDPE, low-density polyethylene; MAH, monoaromatic hydrocarbons; MS, mass spectrometer; PAH, polyaromatic hydrocarbons; PET, polyethylene terephthalate; TGA, thermogravimetric analysis; TPD, temperature programmed desorption.

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precipitation (STRAP) approach as a way to separate the individual polymers layers.⁴ Pyrolysis is a scalable technique which can be used to process multilayer packaging and other forms of mixed plastic waste, without the need for a preliminary separation step.⁵

Pyrolysis is a thermochemical process in which waste plastics are heated in an inert environment and broken down into smaller chemicals which can be of economic value. These chemicals can then in turn be used to produce new plastics such as polypropylene and PET, and thus the process can be referred to as chemical recycling. The chemical recycling of mixed plastic waste via pyrolysis has been found to emit 50% less CO₂ compared to processing the waste via combustion for energy recovery.⁶ Detailed studies on the pyrolysis of mixed plastic waste are valuable as many interactions between various plastic types can occur during their co-pyrolysis.^{7,8} The pyrolysis of mixed plastic waste can be used for the production of pyrolysis oil which can later be fed to a steam cracker for the selective production of target products.⁹ PET is one polymer type not studied often for pyrolysis due its relatively high recycling rate compared to other polymers and the ability to chemically recycle PET via other techniques, such as hydrolysis.¹⁰ Multilayer packages often contain a mixture of both polyolefins and PET.¹¹ Thus, the pyrolysis of PET with and without the presence of polyolefins should be studied for insight on how multilayer packaging can be chemically recycled via pyrolysis. The pyrolysis of PET leads to high yields of benzoic acid, which can deteriorate pyrolysis oil quality, along with causing clogging of reactor lines. The removal of benzoic acid and other heteroatom bound chemicals from pyrolysis oil is an additional processing step that must be taken before the oil can be fed to a steam cracker.^{12,13}

Including zeolite catalysts with plastic waste during its pyrolysis can help plastics to degrade at lower temperatures as well as increasing the yield of target products.¹⁴ The pyrolysis of PET with zeolite catalysts can promote decarboxylation reactions which lead to increased yields of benzene, a target product for the circular economy, and decreased yields of benzoic acid, a problematic chemical for pyrolysis oil and pyrolysis reactor lines.¹⁵ The catalytic pyrolysis of polyolefins can produce high yields of target products such as ethylene, propylene, and butenes (C2-C4 olefins) which can directly be used to produce new plastics. It also can produce high yields of aromatic hydrocarbons such as benzene, toluene, ethylbenzene, and xylene (BTEX) which can serve as platform chemicals for the circular economy. The catalytic pyrolysis of multilayer packaging containing both polyolefins and PET has been shown to produce BTEX yields up to 15 wt%.¹⁶

Previous studies have investigated both the individual pyrolysis of polyolefins and PET, as well as their co-pyrolysis. Xue et al. showed that interactions between PE and PET exist during their co-pyrolysis with HZSM-5 which lead to an increased yield of aromatic products.¹⁷ Our previous work investigated the impact of the zeolite catalyst framework on the interactions which occur between LDPE and PET using thermogravimetric analysis and a micro-pyrolysis GC-MS system.⁸ For the micro-pyrolysis experiments, chemical classes were semi-quantified via a comparison of the MS peak area percentage. The temperature was kept at 500°C and the catalyst:feedstock (CF) ratio was equal to 10 for all of the micro-pyrolysis experiments in our previous work. In this

study, we will build on our previous results by quantifying the weight percent yield of the pyrolysis products utilizing an updated micro-pyrolysis system coupled with two-dimensional gas chromatography (GC × GC), an analytical technique shown to optimize the separation of chemicals found in pyrolysis oils.^{18–20} While synergistic effects between LDPE and PET are known to occur in the presence of zeolite catalysts, the effect of pyrolysis operating conditions on the synergy between these two polymers has not previously been studied. Operating conditions such as pyrolysis temperature and the CF ratio could have an impact on the synergies observed. While the effect of these operating conditions has been studied for the catalytic pyrolysis of polyolefins with zeolite catalysts, they have not been studied for the catalytic pyrolysis of PET. Gaining an understanding on how the operating conditions affect the product yields and the extent of synergistic effects during LDPE and PET catalytic co-pyrolysis, will help us to learn what the ideal operating conditions may be for the co-pyrolysis of these two polymers.

2 | MATERIALS AND METHODS

2.1 | Materials

LDPE pellets were purchased from Sigma-Aldrich and milled into a powder with a SPEX SamplePrep FreezerMill. PET powder was purchased from Goodfellow Corporation (Item number ES306031). Both polymer powders were sieved to a particle size greater than 53 µm and less than 125 µm before use.

ZSM-5 (CBV2314, Si/Al = 11.5) and zeolite Y (CBV600, Si/Al = 2.6) catalysts were supplied from Zeolyst international. The catalysts were calcined in a muffle furnace using a 3°C/min ramp to 550°C and held at 550°C for 5 h in static air. Calcined catalysts were sieved to a particle size greater than 53 µm and less than 125 µm and then stored in a desiccator before use. Silicon carbide powder (200–450 mesh) obtained from Sigma-Aldrich was used as a diluent.

2.2 | Micro-pyrolysis experiments

Catalytic fast pyrolysis experiments were conducted using a Frontier Tandem micro-reactor (Rx-3050TR) connected to a two-dimensional gas chromatography system.

Stainless steel cups, 3.8 mm in diameter and 8 mm in height (Frontier Eco-Cup LF), were used for the pyrolysis experiments. 0.200 ± 0.020 mg of polymer, 0.2–8 mg of zeolite catalyst, and 0–7.8 mg of silicon carbide was added to the cup using an XPR26DR microbalance (Mettler Toledo). The powders were then mixed carefully with a micro-spatula for a period of 5 min, before adding a small layer of quartz wool (3–5 mg). CF ratios of 1, 10, and 40 were used in this study, and determined the weight of zeolite catalyst added to the cup. Silicon carbide was added as a diluent and aided in the collection of solid residuals. The amount of zeolite catalyst and silicon carbide added to the cup depended on the CF ratio which was used. The total weight of catalyst, polymer, and silicon carbide was always equal to 8.2 mg.

A diagram of the micro-reactor for in-situ catalytic pyrolysis experiments can be seen in Figure S3. Prepared sample cups were suspended over a stainless-steel tube (6 mm ID, 120 mm length) surrounded by a heated furnace by use of an autosampler (Frontier Lab). Helium (99.999%, Linde) flow through the furnace was controlled by the GC inlet and kept the furnace pressure at 37 psi. For this study, a split ratio of 270 was used to obtain a flow of 200 mL/min through the reactor. Once the furnace was heated to its set pyrolysis temperature, the pyrolysis cup was dropped into the furnace. Samples in the cup were then instantaneously heated, at an estimated heating rate of 180°C/s.²¹ As pyrolysis volatiles formed, they were swept into the GC inlet and onto the primary column. Once analytes entered the column, they were trapped in the column by use of an MJT-1035E MicroJet Cryotrap (Frontier Lab). The cryotrap was kept on for a period of 5 min after the pyrolysis cup dropped into the reactor at its set pyrolysis temperature. After 5 min, the cryotrap was turned off, allowing products to begin to pass through the primary column. At the same time that the cryotrap was turned off, the reactor was cooled using air flow to a temperature of 200°C, to quench the pyrolysis reactions.

2.3 | GC × GC/FID-MS analysis of micro-pyrolysis volatiles

A two-dimensional gas chromatography (GC × GC) system coupled with a flame ionization detector and a time-of-flight mass spectrometer (BenchTOF, Markes International) was used for the identification and quantification of pyrolysis products. Analytes from the micro-pyrolyzer were first sent through a non-polar primary column (Restek-DH50 50 m × 0.2 mm × 0.5 μm) with a flow rate of 0.76 mL/min. Once analytes passed through the primary column, they then entered a reverse fill-flush modulator (SepSolve Analytical). A modulation period of 2.25 s was used for the GC run time from $t = 0$ –34.5 min before being switched to a modulation period of 3.5 s for the remaining GC run time of $t = 34.5$ –85 min. A flush time of 225 ms was used for both modulation periods. The modulator passed analytes eluting from the primary column onto a polar secondary column (Restek Rxi 17Sil MS 8 m × 0.32 mm × 0.25 μm). A flow rate of 20 mL/min was used for the secondary column. Once analytes exited the secondary column they were fed to a passive splitter (SepSolve Analytical) which split flow to both the MS and FID. Transfer line columns to the MS and FID were cut to dimensions which gave 5 mL/min of flow to the MS and 15 mL/min of flow to the FID at the initial oven temperature of 40°C. The GC oven program used was to hold at 40°C for 10 min, 3°C/min ramp to 110°C, 4°C/min ramp to 300°C, and then hold at 300°C for 5 min.

GC × GC chromatograms from the mass spectrometer were used to identify the pyrolysis products observed. All products identified resulted in a match score above 800 with the NIST library. In addition to the match factor, retention indices reported in the NIST library were also used to aid in the identification. Table S1 provides a complete list of identified products and their corresponding chemical classes. Example GC × GC chromatograms can be seen in Figures S4 and S5.

For the quantification of chemical products observed in the GC × GC FID chromatogram, an external calibration was performed. For the external calibration, laboratory made standards of various concentrations with multiple chemicals (Sigma-Aldrich, >99%) were injected into the front inlet of the GC. The response factor observed for the reference chemicals in the calibration standards were used along with the effective carbon number method to calculate the weight percent yield of the chemical products observed during pyrolysis. A detailed list of the equations used for the GC × GC quantification can be found in the [Supporting Information](#).

Light gas products were quantified using a separate system GC as explained in Section 2.4, and solid residual was quantified as explained in Section 2.5. Combining the yield of products obtained using the various methods led to a mass balance closure between 90 and 110 weight % for pyrolysis experiments with CF ratios between 10 and 40. Experiments with a CF ratio of one led to mass balance closures between 65% and 100%. Lower CF ratios led to an increased formation of non-catalytic products with higher boiling points, thus decreasing the mass balance closure due to the inability for the high boiling point products to elute the GC columns, as has been observed by others.²² All experiments were repeated at least three times. The average yield (wt% and C/O%) of the various chemical products and the associated mass balance closures observed for each experimental condition can be found in Tables S2–S7. The reported yields of the various chemicals and chemical classes are an average of the repeated runs.

2.4 | GC/FID-TCD analysis of micro-pyrolysis volatiles

In order to quantify the light gas products formed during pyrolysis experiments, pyrolysis products were collected in a Tedlar bag (Restek) via the GC split vent, and then subsequently injected into a system GC with FID and TCD detectors (Shimadzu). Ethylene, ethane, propylene, and propane were quantified using a standard FID, while CO and CO₂ were quantified using an FID connected to a methanizer. The TCD detector was used for the detection of nitrogen and thus the presence of air in the gas bag, which could lead to an increase in CO₂ peak area. Further details of the system GC methods and equations used for quantification can be found in the [Supporting Information](#).

2.5 | Solid residual determination via TGA

Once a pyrolysis experiment and its associated GC analysis was complete, the pyrolysis cup was removed from the micro-reactor and stored. Once three pyrolysis cups for a given experimental condition were collected, the used catalyst and silicon carbide for the repeated experiments were combined. The resulting mixture was added to an alumina crucible to be analyzed with an SDT Q600 thermogravimetric analyzer (TA instruments) for temperature programmed oxidation (TPO). During the TPO analysis, the TGA furnace was first heated to 200°C and held for 30 min under 100 mL/min of N₂ (99.999%, Linde) flow to remove any moisture from the catalyst. The carrier gas was

then switched to 100 mL/min of air (99.999%, Linde) flow, and the furnace was heated to 700°C at 10°C/min to burn off the solid residual. The weight change during the air flow step was determined to be the weight of solid residual. The solid residual weight divided by the weight of polymer pyrolyzed for the three pyrolysis experiments was calculated and defined as the solid residual yield. Blank TPO experiments containing fresh catalyst and silicon carbide at the appropriate weights were conducted and applied so that any mass loss due to catalyst degradation or buoyancy effects would not be interpreted as solid residual.

2.6 | Catalyst characterization

A Micromeritics 3Flex surface analysis instrument was used to conduct nitrogen physisorption experiments for the catalysts used in this study. A catalyst mass of 0.12 g was loaded into the analysis tubes and then heated under vacuum to 300°C and held at 300°C for 10 h for degassing. The catalyst mass in the sample tube after degassing was used for the calculations. Analysis was conducted at 77 K and partial pressure (P/P_0) values from $1E-5$ to 0.99 were recorded. The resulting isotherm plots can be seen in Figure S6. Nine P/P_0 values in the range of 0.002–0.01 were used to compute the BET surface area. P/P_0 values in the range of 0.08–0.3, resulting in a thickness range of 0.35–0.5 nm, were used with the Harkins-Jura equation for t-plot analysis to calculate the micropore volume and external surface area. Thickness curves from the t-plot analysis can be seen in Figure S7.

Pyridine was adsorbed onto the zeolite catalysts using the same methodology in our previous work.⁸ The total pyridine loaded onto the zeolite catalyst served as a measure of zeolite acidity and was determined via the temperature programmed desorption (TPD) of pyridine in a thermogravimetric analyzer (TGA). 10 mg of pyridine adsorbed catalyst was added to an alumina crucible which was added to the TGA furnace. 100 mL/min of N_2 (99.999%, Linde) flow was used for the analysis. The TGA furnace was first heated to 200°C and held at 200°C for 30 min to remove any moisture and physisorbed pyridine. The temperature was then ramped to 1000°C using a 10°C/min ramp rate and then held at 1000°C for 10 min to desorb the chemisorbed pyridine. Once the pyridine-TPD run was completed, a blank experiment was run in which catalyst without pyridine adsorbed was subjected to the same TGA method. The weight change from the blank run was subtracted from the pyridine-TPD run, to ensure that any mass loss from catalyst degradation at high temperatures was not being interpreted as pyridine desorption. The total acidity of the catalyst could then be quantified using Equation 1:

$$\text{Acidity} \left[\frac{\text{mmol Pyr}}{g_{\text{cat}}} \right] = \frac{wt_{200^\circ\text{C}} - wt_{1000^\circ\text{C}}}{wt_{t=0} \times fr_{\text{dry cat}}} \times \frac{1 \text{ mmol Pyridine}}{79.1 \text{ mg Pyridine}}, \quad (1)$$

where $wt_{200^\circ\text{C}}$ is the weight of the pyridine loaded catalyst after being held at 200°C for 30 min, $wt_{1000^\circ\text{C}}$ is the recorded weight at the end of the TGA run once all pyridine has been desorbed, $wt_{t=0}$ is the initial

weight of pyridine loaded catalyst which includes water and physisorbed pyridine, and $fr_{\text{dry cat}}$ represents the fraction of pyridine loaded catalyst which is dry catalyst. $fr_{\text{dry cat}}$ is equal to $\frac{wt_{1000^\circ\text{C}}}{wt_{t=0}}$ and is also used to help quantify acidity in the pyridine-FTIR analysis.

Pyridine adsorbed catalysts were analyzed using Fourier transform infrared spectroscopy (FTIR) in transmission mode. 10 mg of pyridine adsorbed catalysts were pressed into supported circular pellets (radius = 0.375 cm, area = 0.442 cm²) by applying 1 ton of pressure for 30 s in a pellet press. The pellet was then analyzed at room temperature using a Bruker Vertex V80 Spectrometer with a LN-MCT detector. Four hundred scans in the 1580–1340 cm⁻¹ range were collected at 6 cm⁻¹ resolution. The resulting spectral band at 1545 cm⁻¹ is representative of pyridine adsorbed to Brønsted acid sites and a band at 1455 cm⁻¹ is representative of pyridine adsorbed to Lewis acid sites. Extinction coefficients of 1.23 and 1.65 cm/μmol were used for the Brønsted acid peak for HZSM-5 and HY catalysts respectively, and an extinction coefficient of 1.87 cm/μmol was used for the Lewis acid peak for both catalysts.²³ The total number of Brønsted or Lewis acid sites for a given catalyst could then be calculated using the modified Beer–Lambert–Bouguer Law:

$$\frac{n^{\text{Pyr}}}{g_{\text{cat}}} = \frac{\text{Peak Area [cm}^{-1}] \times 0.442 \text{ cm}^2}{\epsilon \left[\frac{\text{cm}}{\mu\text{mol}} \right] \times Wt_{\text{cat}} [\text{mg}] \times fr_{\text{dry,cat}}}, \quad (2)$$

where the peak area is the integrated area for the Brønsted or Lewis acid site of interest, 0.442 cm² is the cross-sectional area of the catalyst pellet, ϵ is the extinction coefficient, Wt_{cat} is the weight of pyridine adsorbed catalyst used for the pellet, and $fr_{\text{dry,cat}}$ is the fraction of dry catalyst in the pyridine adsorbed catalyst, as determined by pyridine-TPD.

3 | RESULTS AND DISCUSSION

3.1 | Catalyst characterization

The two zeolite catalysts used in this study were characterized to determine both their surface and acidic properties, and the resulting properties are summarized in Table 1. The BET surface area of the CBV2314 and CBV600 catalysts were determined and observed to be close to the values specified by the supplier (425 and 660 m²/g respectively). The HY catalyst exhibited a larger BET surface area and micropore volume than the HZSM-5 catalyst. The increase in surface area and micropore volume is due to the larger pore diameter (0.74 × 0.74 nm) of the HY catalyst compared to the HZSM-5 catalyst (0.53 × 0.56 and 0.51 × 0.58 nm). The smaller pore diameter of the HZSM-5 is known to limit coke precursor development, leading to lower yields of coke, in comparison with larger pore diameter zeolites such as HY.²⁴ Pyridine-TPD showed that the HZSM-5 catalyst exhibited a higher acidity than the HY catalyst, as almost twice the amount of pyridine was shown to be loaded onto the catalyst (Figure S8). Quantification of the total acid sites using pyridine-FTIR also showed a higher acidity for the HZSM-5 catalyst, as can be seen in Table 1.

TABLE 1 Catalyst characterization results for the two zeolite catalysts used in this study.

Catalyst	BET surface area (m ² /g)	Micropore volume (cm ³ /g)	External surface area (m ² /g)	Acidity TPD (mmol Pyr/g _{cat})	Acidity FTIR (mmol Pyr/g _{cat})	Brønsted: Lewis acid ratio
HZSM-5 (CBV2314)	424	0.134	93	0.843	0.886	35.4
HY (CBV600)	684	0.235	68	0.472	0.316	4.5

This trend of higher acidity for the HZSM-5 catalyst was also observed when quantifying the chemisorbed pyridine using transmission-FTIR. In addition to quantifying the total acidity, the pyridine-FTIR analysis allows for the differentiation of Brønsted and Lewis acid sites (Figure S9). It can be observed that HY exhibits a higher number of Lewis acid sites, and thus a lower Brønsted to Lewis acid ratio. This can be attributed to the CBV600 catalyst exhibiting a higher percentage of extra framework aluminum, which can serve as Lewis acid sites.^{8,25} In Figure S8, higher rates of pyridine desorption are observed at lower temperatures for the HY catalyst, whereas HZSM-5 catalyst has higher rates of desorption at higher temperatures, indicating HZSM-5 has stronger acid sites. A higher acid strength was also observed for HZSM-5 in our previous work, when comparing the NH₃-TPD profiles between the two catalysts.⁸

3.2 | Effect of zeolite type on pyrolysis product distribution

LDPE and PET were pyrolyzed individually with both HZSM-5 and HY catalyst. The pyrolysis temperature was 500°C and the CF ratio was set to 10. The resulting product distribution for LDPE can be seen in Figure 1A, and the resulting product distribution for PET can be seen in Figure 1B.

During the pyrolysis of LDPE, utilizing HZSM-5 as a catalyst led to a higher yield of monoaromatic hydrocarbons (MAH) (21.7 wt% HZ, 4.5 wt% HY) as well as C₂-C₄ olefins (C₂-C₄ O) (21.6 wt% HZ, 11.6 wt% HY). An equal amount of C₂-C₄ paraffins (C₂-C₄ P), and C₅-C₁₀ olefins (C₅-C₁₀ O) were formed for both of the catalysts. HY exhibited a higher yield of C₅-C₁₀ paraffins (C₅-C₁₀ P) (9.0 wt% HZ, 26.3 wt% HY) and a higher yield of solid residual (2.2 wt% HZ, 18.4 wt% HY). The trends observed in our study between the catalysts match well with what had been previously observed in the literature for the catalytic pyrolysis of polyolefins. The continuous pyrolysis of HDPE in a spouted bed reactor has shown that HZSM-5 led to the highest yield of light olefins (58–70 wt%) while HY catalyst led to the highest yield of C₅-C₁₁ non-aromatics (41–47 wt%).²⁶ The shape selective nature of HZSM-5 favors a monomolecular cracking mechanism which leads to high yields of light olefins, while HY favors a bimolecular cracking mechanism, which leads to the formation of isoparaffins and isoolefins.²⁷ This can help to explain the difference in product distribution between the two catalysts. For the in-situ catalytic pyrolysis of PE with HZSM-5 (CBV2314) conducted in a micro-reactor at similar operating conditions (600°C, CF = 20), C₂-C₄ P C% yield was 36.8% while C₂-C₄ O C% yield was 21.2%,¹⁷ closely matching the results obtained in this study.

In our previous study, TGA experiments showed a higher solid residual when LDPE was pyrolyzed with HY (9.7 wt%) than when pyrolyzed with HZSM-5 (0.6 wt%).⁸ Higher coking over HY catalyst than for HZSM-5 catalyst has been observed by others for the catalytic pyrolysis of polyolefins in semi-batch and continuous pyrolysis modes and can be attributed to the larger zeolite framework volume of HY which leads to the buildup of coke precursors.^{24,28}

For PET, it can be seen that HZSM-5 results in a higher yield of benzene (20.8 wt% HZ, 8.0 wt% HY) and CO₂ (36.5 wt% HZ, 20.2 wt% HY), and a lower yield of solid residual (14.7 wt% HZ, 47.6 wt% HY). During PET catalytic pyrolysis, benzoic acid can undergo decarboxylation reactions which lead to the simultaneous formation of benzene and CO₂.²⁹ The formation of benzene via the decarboxylation pathway can be confirmed, as HZSM-5 which shows the highest yield of benzene, also shows a concomitant increase in CO₂ in comparison with HY. The benzene yield for HZSM-5 is similar to the weight% yield observed by others (25.6%) for PET/HZSM-5 pyrolysis in a micro-reactor at similar operating conditions (600°C, CF = 20).¹⁷ A higher yield of CO (6.7 wt% HZ, 8.8 wt% HY) and acetaldehyde (0.5 wt% HZ, 4.1 wt% HY) was observed for the HY catalyst, which shows its propensity to favor the decarbonylation pathway over the decarboxylation pathway.⁸ The higher solid residual over HY catalyst for PET pyrolysis is similar to the trend that was seen during LDPE pyrolysis. In our previous work, HY resulted in a solid residual% of 29.2% while HZSM-5 resulted in a solid residual of 17.4% for PET pyrolysis in a TGA at 1:1 CF ratio.⁸ The higher solid residual of 49% observed for PET/HY in this study could be due to the higher CF ratio of 10 which was used.

The yield of chemical product classes for the catalytic co-pyrolysis of LDPE and PET at a 1:1 LDPE:PET ratio can be seen in Figure 2A. HZSM-5 outperformed HY for the production of C₂-C₄ olefins (12.2 wt% HZ, 8.0 wt% HY), benzene (9.1 wt% HZ, 3.1 wt% HY), and alkylated benzenes (13.5 wt% HZ, 6.4 wt% HY). These three classes of chemicals can all be used in the production of new plastics, contributing to the circular economy. In total, HZSM-5 led to almost double the yield of these products (34.8 wt%) in comparison with HY (17.5 wt%). Similar findings have been observed during the catalytic co-pyrolysis of PE and PET containing multilayer packaging, in which the BTEX yield for HZSM-5 was 14.9 wt% and for HY was 8.1 wt%.¹⁶ HY produced lower yields of the beforementioned target products due to its propensity to form high yields of C₅-C₁₀ paraffins (9.9 wt%) and solid residual (40.1 wt%).

In addition to evaluating the product yields for the 1:1 co-pyrolysis of LDPE and PET with the two catalysts, insights can also be made on the synergistic effects that occur between these two polymers during their co-pyrolysis. An additive yield of product

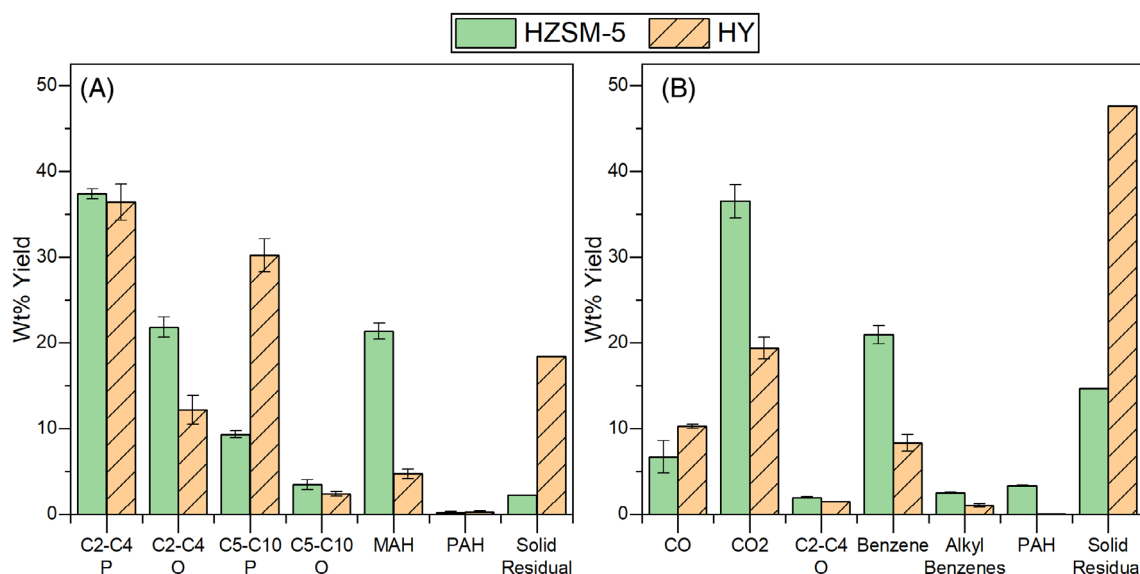


FIGURE 1 Effect of zeolite catalyst on the individual pyrolysis of LDPE (A) and PET (B) for catalyst:feedstock ratio = 10 and pyrolysis temperature = 500°C.

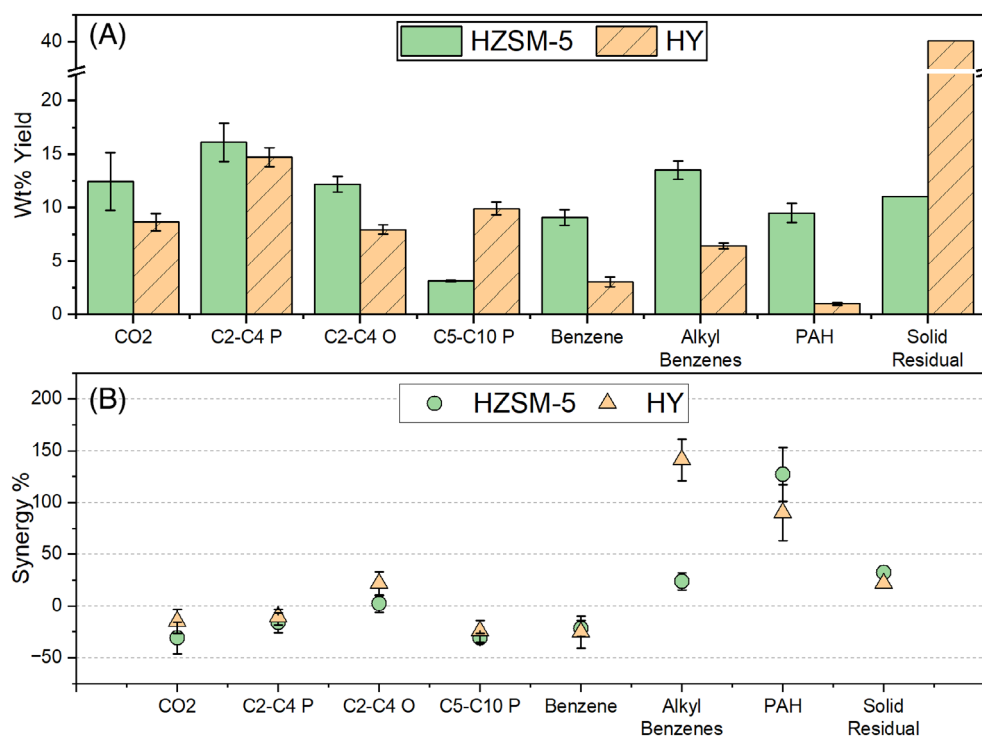


FIGURE 2 Effect of zeolite catalyst type on product yield (A) and synergy (B) for catalytic co-pyrolysis of LDPE and PET at a 1:1 LDPE:PET ratio, 10:1 catalyst:feedstock ratio, and pyrolysis temperature of 500°C.

classes were calculated, based on the yields observed for the individual pyrolysis of the polymers, using Equation 3.

$$\text{Additive yield (x)} = \frac{w_{\text{LDPE}}}{w_{\text{LDPE}} + w_{\text{PET}}} \times \text{LDPE yield (x)} + \frac{w_{\text{PET}}}{w_{\text{LDPE}} + w_{\text{PET}}} \times \text{PET yield (x)} \quad (3)$$

where w_{LDPE} and w_{PET} are the weights of LDPE and PET added to the micro-pyrolysis cup for a given co-pyrolysis experiment, LDPE yield (x) is the yield of chemical x for the individual pyrolysis of LDPE at the given operating condition, and PET yield (x) is the yield of chemical x for the individual pyrolysis of PET. Comparing the additive yield with the actual yield observed during co-pyrolysis allows for a

calculation for a synergy% as shown in Equation 4. The synergy% that was observed for the two catalysts can be seen in Figure 2B.

$$\text{Synergy\%(x)} = \frac{\text{Actual yield (x)} - \text{Additive yield (x)}}{\text{Additive yield (x)}} \times 100\%. \quad (4)$$

For both zeolite catalysts, certain chemical classes were observed to have either a positive or negative synergy. A negative synergy was observed for CO₂ for both HZSM-5 (−31%) and HY (−15%), with a slightly stronger synergy observed for the HZSM-5 catalyst. C2-C4 paraffins showed a negative synergy (−16% HZ, −11% HY), while interestingly C2-C4 olefins showed a neutral to positive synergy for the two catalysts (2% HZ, 22% HY). A negative synergy was seen for both C5-C10 paraffins (−31% HZ, −24% HY) and C5-C10 olefins (−28% HZ, −17% HY). Benzene resulted in a negative synergy (−22% HZ, −25% HY), while alkyl benzenes (24% HZ, 141% HY), polyaromatics (127% HZ, 90% HY), and solid residual (32% HZ, 22% HY) all resulted in a positive synergy. While the direction of the synergy for all the classes was independent of the catalyst used, the most noticeable difference in the magnitude of synergy between the two catalysts was seen for the alkyl benzenes. For the catalytic co-pyrolysis of cellulose and polyethylene with HZSM-5 and SAPO-34 catalysts, it was shown that the addition of iron into the HZSM-5 catalyst decreased the Brønsted to Lewis acid ratio, and increased the synergy observed toward alkylated benzenes.³⁰ Also, decreasing the Brønsted to Lewis acid ratio through the addition of nickel to ZSM-5 catalyst has been shown to increase the extent of aromatization reactions which produce alkylated benzene products during the catalytic pyrolysis of polypropylene.³¹ In our study, HY was observed to have a lower Brønsted to Lewis acid ratio than for HZSM-5, and thus we believe this may be the reason HY was observed to have a higher synergy toward alkylbenzenes. It has also been observed that when using a 10:1 CF ratio at 500°C, HY resulted in a higher synergy toward aromatic hydrocarbons than for HZSM-5 catalyst during the catalytic co-pyrolysis of LLDPE and cellulose, and this was attributed to a higher diffusion efficiency into the HY pores by the authors.³² The higher diffusion efficiency into HY pores can offer a secondary explanation for the increased synergy. Benzene content in gasoline needs to be minimized, while alkylated benzene content is desirable, as these chemicals help to serve as octane boosters for fuel. Thus, the positive synergy for alkyl benzenes and the negative synergy for benzene which is observed for HY is desirable, as this catalyst is also producing the largest yield of C5-C10 paraffins. The very large yield of solid residual for the HY catalyst is however, undesirable. Due to high yield of chemicals valuable for the circular economy, along with the low solid residual yield observed, HZSM-5 was selected as the catalyst to be used when investigating the effects of the pyrolysis operating conditions.

3.3 | Effect of CF ratio

LDPE and PET were pyrolyzed individually with HZSM-5 catalyst at various CF ratios using a pyrolysis temperature of 500°C. The

resulting product distribution for LDPE can be seen in Figure 3A and the product distribution for PET can be seen in Figure 3B.

For LDPE, decreasing the CF ratio from 40 to 1 led to four times the yield of C2-C4 olefins (56.5 wt% CF = 1, 14.0 wt% CF = 40). In addition to C2-C4 olefins, lowering the CF ratio also increased the yield of C5-C10 olefins (12.4 wt% CF = 1, 0.9 wt% CF = 40). Increasing the CF ratio from 1 to 40 led to an increase in C2-C4 paraffins (11.6 wt% CF = 1, 50.5 wt% CF = 40), monoaromatic hydrocarbons (7.4 wt% CF = 1, 21.4 wt% CF = 40), and solid residual (0 wt% CF = 1, 7.8 wt% CF = 40). This trend of increased light olefin yield for lower CF ratios is one that has been observed by others. It was shown using a lab scale batch reactor (1.6 g polyolefin plastic) with nanosized HZSM-5, that increasing the CF ratio from 0.01 to 0.25 led to a decrease in the selectivity to C2-C4 olefins (34%–19%) and in increase in the selectivity to C1-C4 paraffins (10%–27%) and C6-C9 aromatics (4%–23%).³³ At low CF ratios, the end chain scission of LDPE with HZSM-5 occurs, forming high yields of C2-C4 olefins. Increasing the CF ratio encourages secondary reactions with the HZSM-5 catalyst to occur, leading to the oligomerization of C2-C4 olefins to form aromatic hydrocarbons. As aromatization reactions occur, hydrogen is released, leading to the hydrogenation of C2-C4 olefins to form increased levels of C2-C4 paraffins.³³ For the ex-situ catalytic pyrolysis of polyolefins with HZSM-5, it has been shown that increasing the catalyst contact time, thus increasing the extent of secondary reactions, led to an increase in aromatics production and a decrease in the production of C2-C4 olefins.²² Similar effects were observed for the conversion of 1-octene as a plastic waste model compound with HZSM-5, in that longer contact times led to increased BTX and C2-C3 paraffins yield, and decreased C2-C5 olefins yield.³⁴ It has been previously reported that the ex-situ pyrolysis of PE with HZSM-5 led to a higher yield of C2-C4 olefins (80.3 C%) than for in-situ pyrolysis (20.2 C%) when using a CF ratio of 20 and a pyrolysis temperature of 600°C.¹⁷ Our findings show that by decreasing the CF ratio for in-situ pyrolysis at lower reaction temperatures of 500°C, the C2-C4 olefins yield increase and begin to approach the level that has been obtained by others for ex-situ pyrolysis.^{35,36} By decreasing the CF ratio, the possibility for secondary reactions on the catalyst which can convert C₂-C₄ olefins into C₂-C₄ paraffins and aromatics is decreased. For many ex-situ pyrolysis studies, a micro-reactor with high flow rates is used which limits the contact time for pyrolysis volatiles with the zeolite catalyst.^{17,37} When flow rates are decreased and contact time is increased, the yield of C₂-C₄ olefins decrease while the yield of aromatics increase for HZSM-5 catalyst.²² For in-situ catalytic pyrolysis in a micro-reactor, the contact time of pyrolysis volatiles with the catalyst may be increased, due to mass transfer limitations which occur in the pyrolysis cup used for the Frontier micropyrolyzer.²¹ Thus to limit the secondary reactions for in-situ pyrolysis in the micro-reactor system, lower CF ratios may need to be used in comparison with those used for ex-situ pyrolysis.

For PET it can be seen that increasing the CF ratio leads to an increase in the yield of benzene, CO₂, polyaromatics, and solid residual. A large difference in the benzene yield is observed when the CF ratio is increased from 1 (2.4 wt%) to 10 (21.0 wt%). For CF equal to 1, the yield of oxygenated products (14.0 wt%) is much higher than

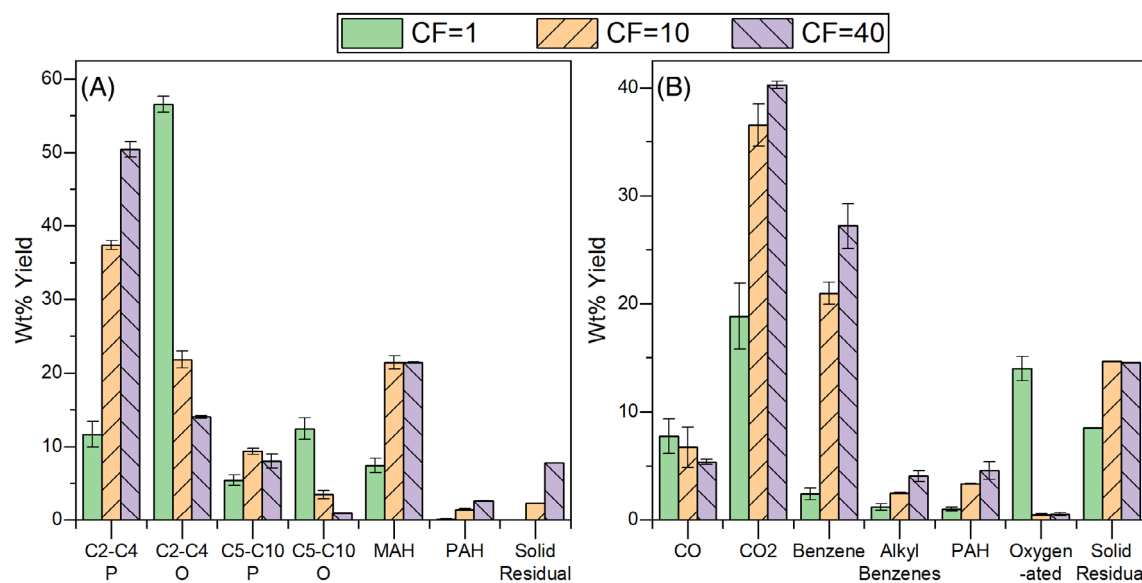


FIGURE 3 Effect of the catalyst:feedstock (CF) ratio on the product distribution for LDPE (A) and PET (B) with HZSM-5 catalyst at 500°C.

for the higher CF ratios of 10 and 40 (0.5 wt% CF = 10, 0.5 wt% CF = 40). Higher CF ratios provide additional catalyst to aid in the deoxygenation reactions which lead to higher yields of benzene and lower yields of oxygenated products. For CF equal to 1, the oxygenated product with the highest yield is benzoic acid (8.1 wt%). Benzoic acid is known to be a major product for the non-catalytic pyrolysis of PET.³⁸ Thermogravimetric analysis has shown that for PET/HZSM-5 at a CF ratio of 1, a mixture of catalytic and non-catalytic degradation occurs,⁸ and thus observing a mixture of catalytic products such as benzene and non-catalytic products such as benzoic acid is logical. Benzoic acid is an undesirable product as its corrosiveness can deteriorate the quality of pyrolysis oil, and can clog pipes, causing serious problems at the industrial scale.³⁹ Thus, the complete conversion of benzoic acid and other oxygenated products to benzene and other hydrocarbons is desired. For this reason, CF ratios higher than 1 should be used for PET.

The effect of CF ratio on the yield of pyrolysis products for the 1:1 mixture of LDPE and PET at 500°C can be seen in Figure 4A with the associated synergies seen in Figure 4B. Utilizing a CF ratio of 1 led to the largest yield of C2-C4 olefins (22.2 wt%), C5-C10 paraffins and olefins (6.2 wt%), and benzoic acid (2.6 wt%). Increasing the CF ratio led to an increase of all other product classes namely CO₂, C2-C4 paraffins, benzene, alkyl benzenes, polyaromatics, and solid residual. Increasing the CF ratio from 10 to 40 had a minor impact on the product distribution ($p = 0.285 \pm 0.152$) where increasing the CF ratio from 1 to 10 had a major impact ($p = 0.023 \pm 0.035$). Utilizing a lower CF ratio helps to increase the yield of C2-C4 olefins from LDPE by minimizing secondary reactions but leads to a decrease in the yield of benzene from PET due to the lack of catalyst needed for the complete decarboxylation of benzoic acid. Thus, the CF ratio of 1 which is ideal for LDPE, is not ideal for PET, and the CF ratios of 10–40 which are ideal for PET are not ideal for LDPE.

The CF ratio which is used for the co-pyrolysis of LDPE and PET is shown to have a direct impact on the level of synergies which are observed. At a CF ratio of 10, the synergy for benzene is negative due to benzene formed from PET reacting with hydrocarbon intermediates from LDPE to form increased levels of alkylated benzenes, polyaromatics, and coke. When the CF ratio is equal to 1, the synergy for benzene becomes positive, concomitant with a negative synergy for benzoic acid. It is theorized that the presence of LDPE helps to decarboxylate benzoic acid by supplying an additional source of hydrogen which aids the deoxygenation reactions. It has been shown that co-pyrolyzing polypropylene plastic alongside biomass with HZSM-5 helps to increase deoxygenation leading to a bio-oil with lower oxygen content compared to when pyrolyzing biomass alone.⁴⁰ The presence of hydrogen has been shown to promote the catalytic decarboxylation of various carboxylic acids.^{41,42} For the catalytic pyrolysis of PET with HZSM-5, it has been shown the conducting the pyrolysis in H₂ atmosphere increased the yield of benzene compared to catalytic pyrolysis conducted in an inert atmosphere.¹⁷ The catalytic pyrolysis of polyolefins such as HDPE and PP with ZSM-5 has been shown to produce hydrogen gas.⁴³ Thus it is believed that the release of hydrogen which would occur during the catalytic pyrolysis of LDPE, is aiding in the decarboxylation of benzoic acid, formed during the catalytic pyrolysis of PET. Along with a positive benzene synergy, using a lower CF ratio of 1 leads to a negative synergy for C2-C4 olefins, which is different than for the other CF ratios. It has been shown through thermogravimetric analysis a competition between LDPE and PET during co-pyrolysis with HZSM-5 at a CF ratio of 1, leading to a decreased level of low temperature degradation for LDPE.⁸ Increasing the CF ratio leads to an increase in the synergy for C2-C4 olefins. No synergy toward C2-C4 olefins was observed for a CF ratio of 10 and a positive synergy for C2-C4 olefins can be seen for the CF ratio of 40. Increasing the CF ratio

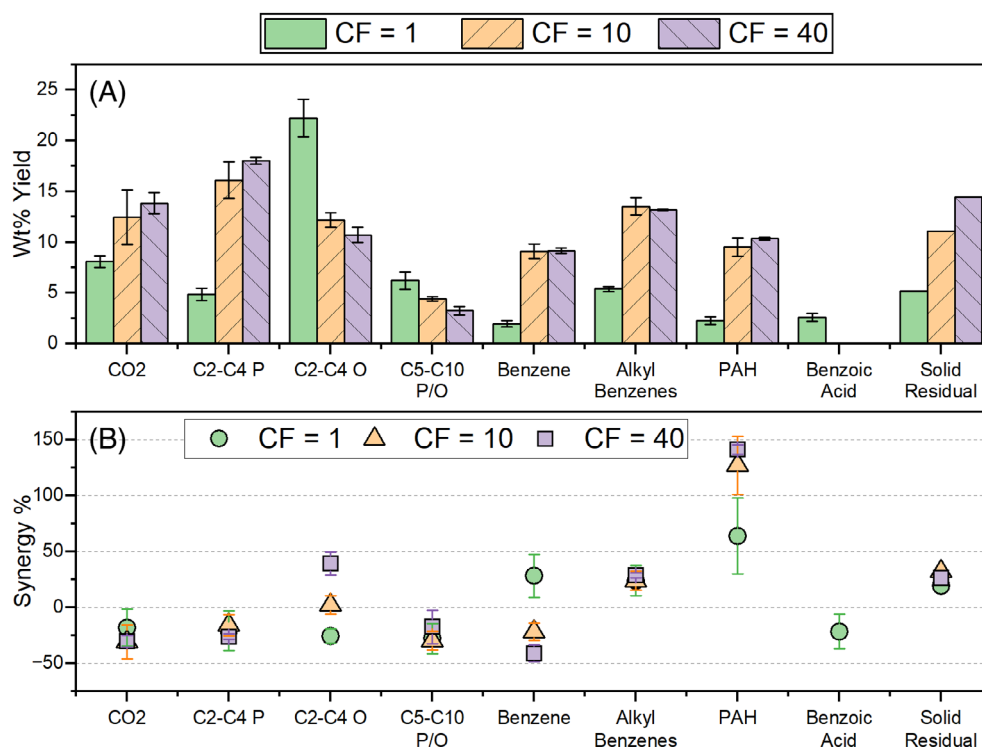


FIGURE 4 Effect of catalyst:feedstock (CF) ratio on the product yields (A) and synergistic effect (B) for the catalytic co-pyrolysis of LDPE and PET (1:1 LDPE:PET ratio, temperature = 500°C, HZSM-5).

also led to an increase in the synergy toward polyaromatics. Regardless of the CF ratio, the synergy toward CO₂, C2-C4 paraffins, C5-C10 paraffins, alkyl benzenes, and solid residual stayed the same.

3.4 | Effect of pyrolysis temperature

LDPE and PET were pyrolyzed individually with HZSM-5 catalyst at a CF ratio of 10, using various pyrolysis temperatures. The resulting product distribution for LDPE can be seen in Figure 5A and the product distribution for PET can be seen in Figure 5B.

Figure 5A shows the effect of pyrolysis temperature for the product distribution of LDPE. It can be seen that varying the pyrolysis temperature from 400°C to 600°C does not have a substantial effect on the product distribution. While the effect appears to be minimal, some trends can be seen. As the temperature increased from 400°C to 600°C the yield of C5-C10 paraffins decreased from 11.7 to 7.3 wt%, and the yield of monoaromatic hydrocarbons increased from 18.7 to 22.8 wt%. Similar trends were observed for the continuous catalytic pyrolysis of HDPE with HZSM-5, during which the monoaromatics yield increased from 9.1 to 20.7 wt% and the nonaromatic C5-C11 yield decreased from 16.6 to 11.5 wt% when the temperature was increased from 450°C to 570°C.⁴⁴ While increasing temperature has been shown to decrease C2-C4 olefin yield for the catalytic pyrolysis of polyolefins,³³ for our study, the C2-C4 olefin yield remained constant. This could be due to the

higher CF ratio of 10 which was used when the temperature was varied, a CF ratio in which the C2-C4 olefin yield is not at its maximum.

In Figure 5B, the effect of pyrolysis temperature on the major products from the pyrolysis of PET can be seen. It is shown that the pyrolysis temperature has a more dramatic effect on the product distribution for PET than for LDPE. A higher yield of benzene, alkyl benzenes, and polyaromatics is seen as the temperature is increased and a decrease in the solid residual is observed. A large solid residual yield of 38.6 wt% is observed at 400°C. The degradation of PET in the presence of HZSM-5 is known to occur in the temperature range of 350°C–500°C.⁸ 400°C is too low of a pyrolysis temperature for the complete conversion of PET, leading to a high solid residual which is likely a blend of both unreacted polymer, and catalytic coke. The derivative TPO curve for the 400°C used catalyst exhibits an additional low temperature peak at 420°C which is not present for the used catalyst collected from the 500°C or 600°C pyrolysis experiments (Figure S10). The mass loss peak at 420°C can be interpreted as unreacted polymer, where the mass loss peak at 550°C observed for all the used catalysts can be interpreted as catalytic coke. Increasing the temperature from 400°C to 500°C allows for a more complete conversion of PET leading to a decrease in the solid residual yield from 38.6 to 14.7 wt% and an increase in both the benzene yield from 13.5 to 20.8 wt% and the CO₂ yield from 24.5 to 36.5 wt%.

The product distribution for the catalytic co-pyrolysis of LDPE and PET (1:1 wt%) at various temperatures can be seen in Figure 6A.

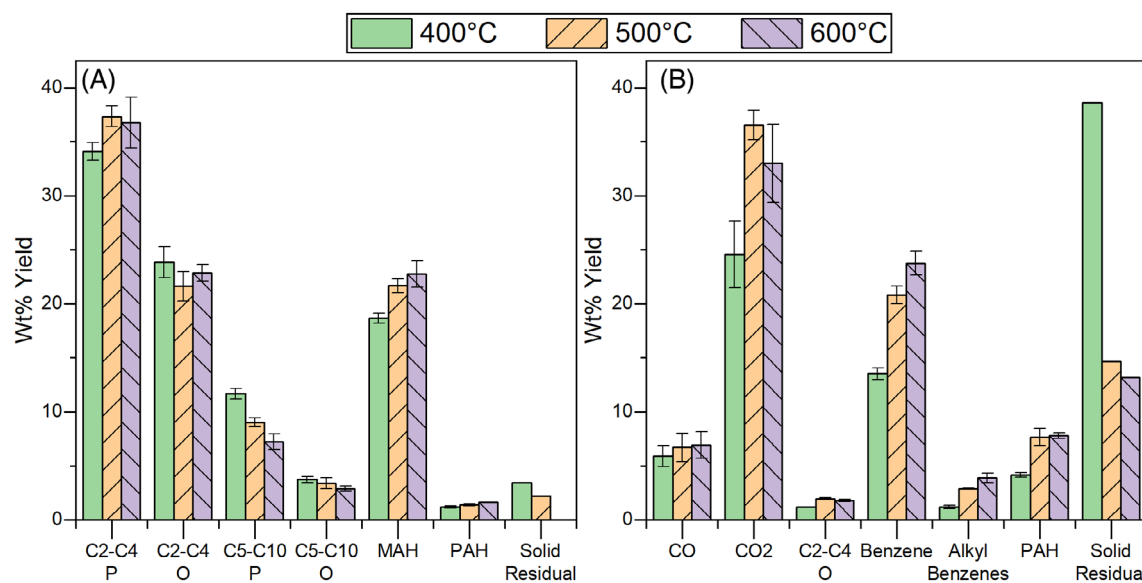


FIGURE 5 Effect of pyrolysis temperature on product yields for the catalytic pyrolysis of LPDE (A) and PET (B) with HZSM-5 catalyst for a 10:1 catalyst:feedstock ratio.

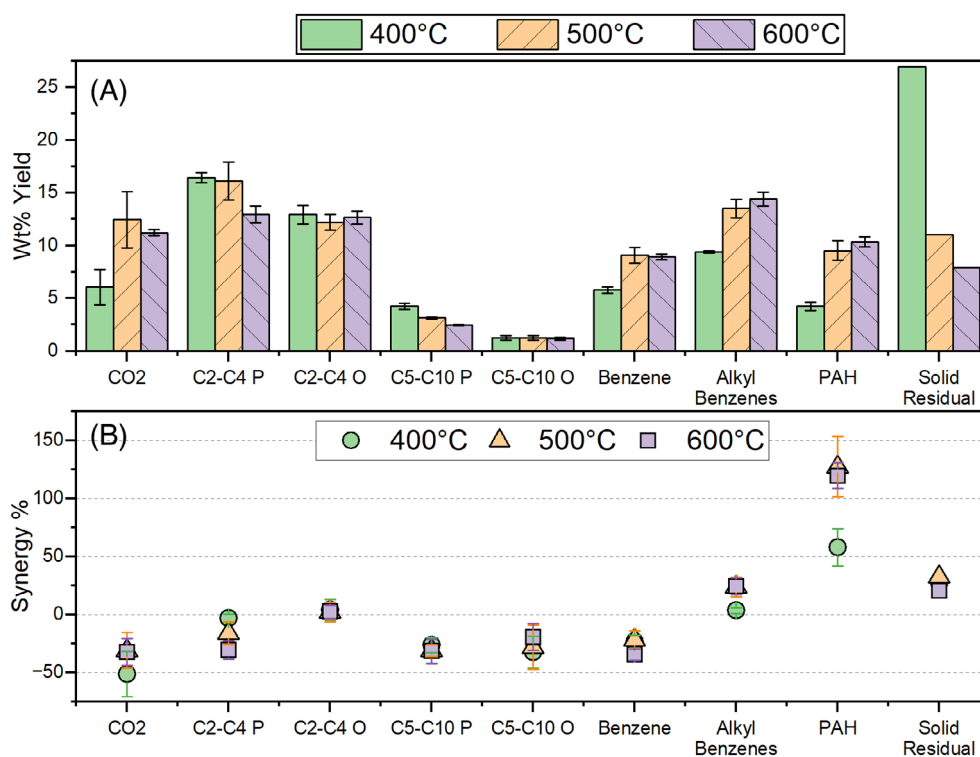


FIGURE 6 Effect of pyrolysis temperature on the product yields (A) and associated synergies (B) observed for the catalytic co-pyrolysis of LDPE and PET at a 1:1 LDPE:PET ratio, and a 10:1 catalyst:feedstock ratio.

Increasing the temperature did not have any impact on the production of C2-C4 olefins, which matches with what was observed for the individual pyrolysis of LDPE. Increasing the temperature did however lead to an increase in benzene, alkyl benzenes, and polyaromatics and a decrease in the solid residual, which are similar trends observed for the individual pyrolysis of PET. The same low temperature TPO peak

observed for PET at 400°C was also observed for LDPE and PET at 400°C, indicating the incomplete conversion of PET during the co-pyrolysis (Figure S11). While in our micro-pyrolysis experiments a temperature of 400°C seemed sufficient for LDPE, others who have investigated the continuous catalytic pyrolysis of polyolefins at larger scale, have recommend a minimum temperature of 500°C, in order to

avoid operational problems which occur from the accumulation of waxes in the reactor at lower temperatures.⁴⁴ This suggestion along with our results showing incomplete conversion of PET at 400°C, demonstrate 500°C to be a more ideal pyrolysis temperature than 400°C, even when a catalyst is involved.

Slightly higher levels of synergies for the various chemical classes are observed as the temperature increases. The temperature appears to affect the level of synergy most for the C2-C4 paraffins, alkyl benzenes, and the polyaromatic hydrocarbons. Increasing the pyrolysis temperature from 400°C to 600°C decreased the C2-C4 paraffin synergy from −3% to −31%, increased the alkylated benzene synergy from 3% to 24%, and increased the polyaromatic synergy from 58% to 120%. For the co-pyrolysis of biomass with plastic waste over HZSM-5, it has been shown that increasing the pyrolysis temperature from 500°C to 600°C led to a more positive synergy for the formation of aromatic hydrocarbons.³²

3.5 | Effect of LDPE:PET ratio

The catalytic co-pyrolysis of LDPE and PET was conducted at additional LDPE:PET ratios of 9:1 (10% PET) and 3:1 (25% PET) for a pyrolysis temperature of 500°C and a CF ratio of 10. The resulting product distribution and synergies can be seen in Figure 7. It can be observed that as higher percentages of PET are added to the mixture, lower yields of C2-C10 paraffins and olefins and alkyl benzenes are

observed. Increasing PET content increased the CO₂, benzene, polyaromatic, and solid residual yield.

An interesting effect of the LDPE:PET ratio on the synergies observed for some of the product classes is seen. Most notably, it can be seen that the solid residual synergy increases for lower percentages of PET. The solid residual yield for 10% PET was 7.8 wt% and for 25% PET was 8.7%. Increasing the PET content by 15% (10%–25%) only led to a 0.9% increase in the solid residual. The opposite trend is observed for the *magnitude* of synergy for polyaromatics, alkylated benzenes and C2-C10 paraffins. The *magnitude* of the synergy for these chemical classes increased as higher percentages of PET were used. For example, the synergy for polyaromatics increased from 52% to 127% and the synergy for C5-C10 paraffins decreased from −8% to −31%.

3.6 | Surface response analysis

Response surface methodology (RSM) is a technique which can be used to evaluate the effect of various input variables on a response factor of interest, as well as for finding which input parameters should be used to optimize the response. RSM has been used to find the optimal pyrolysis operating conditions and catalyst properties for maximizing the yield of pyrolysis oil and monoaromatic hydrocarbons.^{45,46} Response surface regression was conducted in Minitab for the pyrolysis experiments conducted with HZSM-5 at various CF ratios, temperatures, and LDPE:PET ratios. Response surface

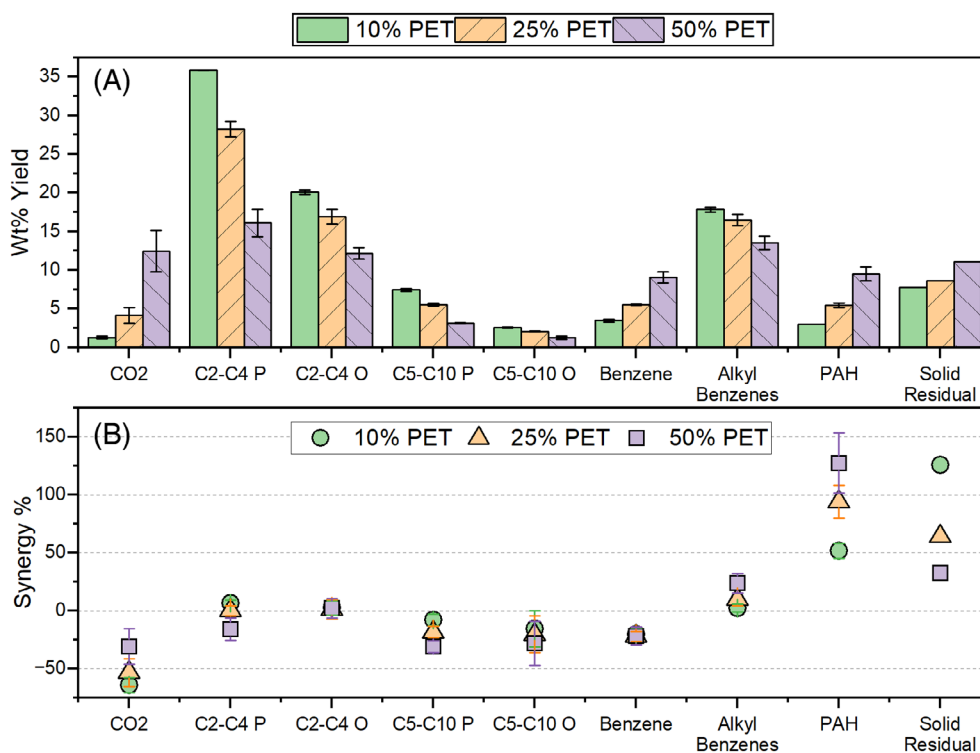


FIGURE 7 Effect of LDPE:PET ratio on the product yields (A) and associated synergies (B) for the catalytic co-pyrolysis of LDPE and PET with HZSM-5 at a temperature of 500°C and a CF ratio of 10:1.

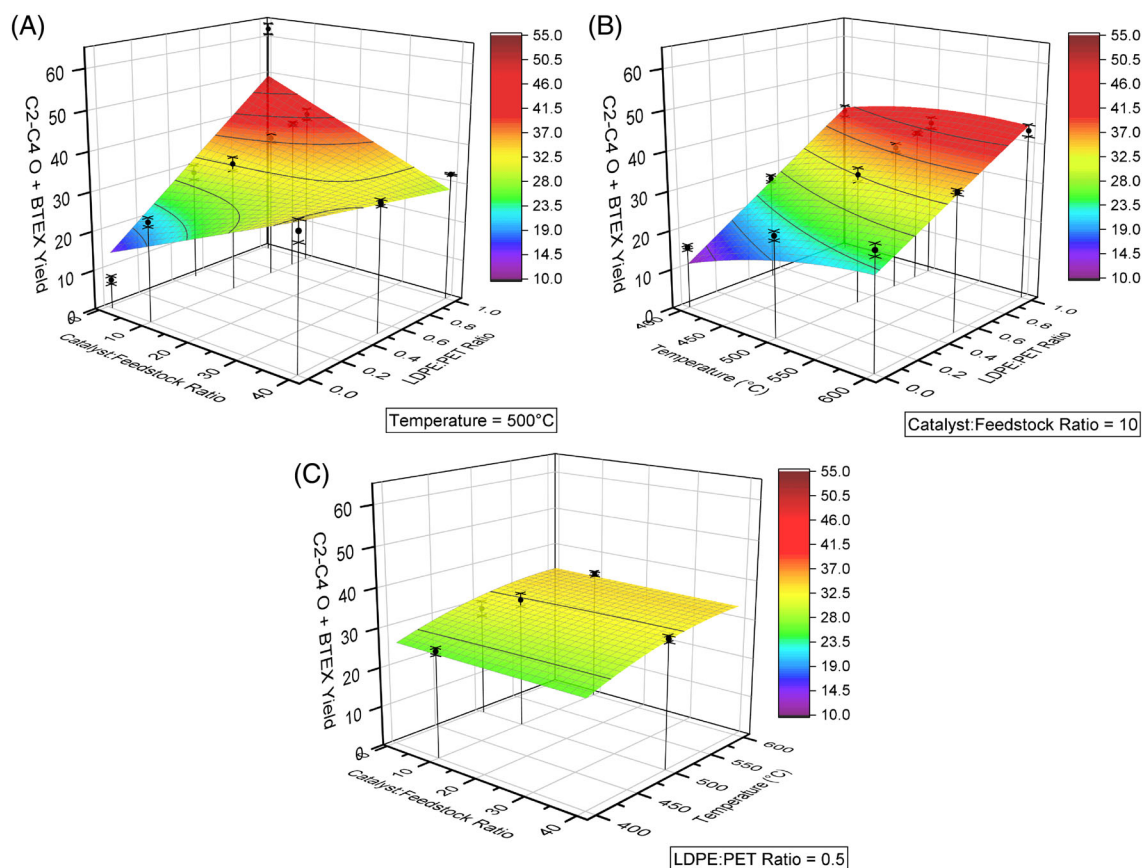


FIGURE 8 Surface response curves demonstrating the effect of operating conditions on the yield of C2-C4 olefins and BTEX chemicals for the catalytic co-pyrolysis of LDPE and PET with HZSM-5 catalyst for constant temperature (A), constant catalyst:feedstock ratio (B), and constant LDPE:PET ratio (C).

regression was conducted for both the yield of C2-C4 olefins and yield of BTEX. Table S13 lists the experiments conducted, along with their observed response yields.

The resulting polynomial regression equations along with their associated statistical details can be found in the [Supporting Information](#). The resulting polynomial equations were combined to express the yield of C2-C4 olefins and BTEX and were plotted along with experimental data in Figure 8.

Figure 8A shows the effect of CF ratio and LDPE:PET ratio on the yield of C2-C4 olefins and BTEX for a pyrolysis temperature of 500°C. The maximum yield can be observed for a CF ratio and an LDPE:PET ratio both equal to 1. When the CF ratio is held constant at 1, there is a sharp decrease in the yield as the LDPE:PET ratio is decreased to zero. In other words, for low CF ratios, the LDPE:PET ratio makes a large difference on the yield of valuable products. A different trend is observed however when the CF ratio is increased to 40. It can be seen that for a CF ratio of 40, decreasing the LDPE:PET ratio does not have an impact on the C2-C4 O + BTEX yield. Figure 8B shows the effect of temperature and LDPE:PET ratio on the C2-C4 O + BTEX yield for a CF ratio of 10. For all of the temperatures tested, it is seen that increasing the LDPE:PET ratio leads to an increase in the C2-C4 O + BTEX yield. While increasing the

temperature does improve the yield of target products at lower LDPE:PET ratios, at higher LDPE:PET ratios, increasing the temperature makes less of an impact. Figure 8C shows the effect of CF ratio and temperature for a constant LDPE:PET ratio of 0.5. There is less curvature in this plot, indicating that at this LDPE:PET ratio, varying the operating conditions does not make as much of an impact on the yield of target products. This is because the CF ratios which are ideal for LDPE are not ideal for PET, and vice versa, as discussed in Section 3.3. At lower CF ratios where the yield of C2-C4 olefins from LDPE are maximized, the conversion of benzoic acid from PET to benzene is minimized. For higher CF ratios where benzoic acid is completely converted to benzene, the C2-C4 olefin yield is decreased due to the conversion to C2-C4 paraffins.

3.7 | Data availability and reproducibility statement

Product yields for the various pyrolysis experiments which are shown in Figures 1–7 are tabulated in Tables S2, S4, and S6. The error bars shown for the product yields in Figures 1–7 show the standard deviation for repeated experiments. All experimental conditions were

repeated at least three times. The solid residual yield does not contain error bars as only one TPO analysis was performed for a given experimental condition, as is explained in Section 2.5. The synergy percents reported in Figures 2, 4, 6, and 7 are tabulated in Tables S8–S11. The error bars for the synergy% were calculated following formulas for error propagation. Having another researcher repeat key steps from an experiment is one way to ensure experimental reproducibility.⁴⁷ All steps for a given pyrolysis experiment were repeated in triplicate by two different researchers in our group. A comparison of the results between the two researchers can be found in Table S12.

4 | CONCLUSION

The catalytic (co-)pyrolysis of LDPE and PET with HY and HZSM-5 catalysts at different CF ratios, pyrolysis temperatures, and LDPE:PET ratios was conducted using a micro-pyrolysis two-dimensional gas chromatography system. A new instrumental method of split gas capture was used and described for the micro-pyrolysis system, which allowed for the separation and quantification of important light gas products (propylene, propane, CO₂, etc.). For the first time, an investigation into how pyrolysis operating conditions affected the nature and magnitude of synergistic effects occurring between LDPE and PET during their co-pyrolysis with zeolite catalysts was conducted. HY catalysts resulted in a synergistic effect which led to a 141% increase in the yield of alkylated benzenes for the co-pyrolysis of LDPE and PET from what was expected from the pyrolysis of the individual polymers. HZSM-5 however showed a much lower yield of solid residual (11 wt%) than for HY (40 wt%) as well as an increase in the yield of desirable products such as C2–C4 olefins and BTEX (35 wt%). Decreasing the CF ratio from 10 to 1 for LDPE/HZSM-5 pyrolysis led to a drastic increase in the yield of C2–C4 olefins (22 to 57 wt%) but led to a decrease in the yield of benzene for PET/HZSM-5 pyrolysis (21 to 2 wt%) with a subsequent increase in the yield of benzoic acid (0 to 8.1 wt%). Utilizing a lower pyrolysis temperature of 400°C was sufficient for the pyrolysis of LDPE but led to the incomplete conversion of PET. To the author's knowledge this is the first time a study has shown that operating conditions which are most ideal for the catalytic pyrolysis of LDPE, are not ideal for the catalytic pyrolysis of PET. For the co-pyrolysis of LDPE and PET at a CF ratio of 1, a synergistic effect was observed which led to a 22% decrease in benzoic acid yield and a 28% increase in benzene yield from what was predicted from the individual pyrolysis experiments, indicating that the presence of LDPE was beneficial for the catalytic pyrolysis of PET at low CF ratios. Surface response diagrams were created to allow for a visualization of the effect of multiple operating conditions on the yield of target products such as C2–C4 olefins and BTEX.

AUTHOR CONTRIBUTIONS

Sean Timothy Okonsky: Conceptualization; writing – original draft; writing – review and editing; visualization; validation; methodology; investigation; software; formal analysis; data curation. **Neil Robert Hogan:** Investigation; data curation. **Hilal Ezgi Toraman:**

Conceptualization; funding acquisition; writing – review and editing; visualization; methodology; project administration; resources; supervision.

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DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available in the supplementary material of this article.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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